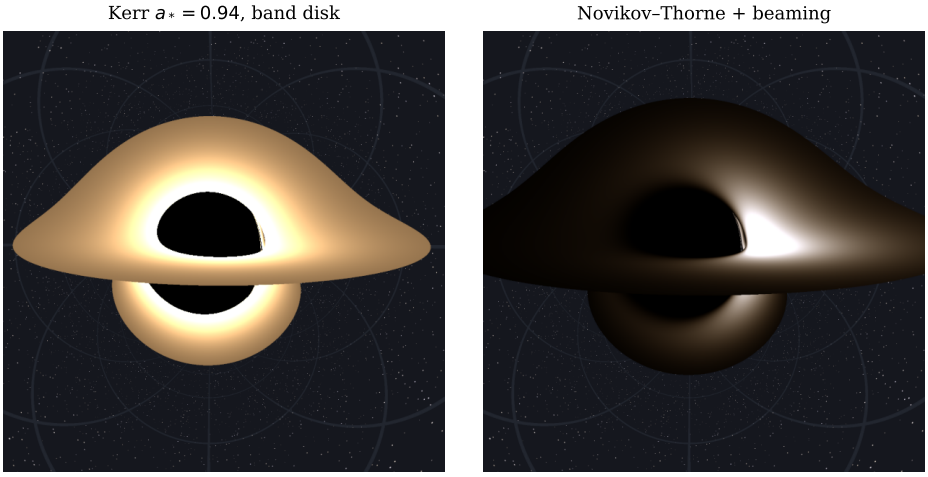
SGR A*

A First-Principles Relativistic Simulator of the Galactic-Centre Black Hole

Shadow metrology, S-star orbit fitting, and a free-quadrupole test of the no-hair theorem — with a complete account of the methods, tests, and development of the instrument



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Independent research report

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Abstract

We present a ground-up, physically rigorous simulator of the Galactic-centre black hole Sgr A*, built on exact Kerr geodesics and validated end-to-end against published strong-field observations. Null and timelike geodesics are integrated in a second-order Mino-time formulation of the Carter equations with per-step constraint projection, conserving (E, L_z, Q) to better than 10^{-6} relative over complete orbits. A CPU reference ray tracer reproduces the analytic Schwarzschild shadow to a relative accuracy of 5×10^{-16} and the Bardeen critical curve for spinning holes; a double-precision GPU compute port agrees with the CPU ground truth to a mean pixel residual of 0.00289. Fitting 13 orbital and potential parameters to the 145 astrometric and 44 radial-velocity measurements of the star S2 (Gillessen et al., 2017) with full general-relativistic timelike geodesics recovers $M = 4.270 \pm 0.195 \times 10^6\,\mathrm{M}_\odot$ and $R_0 = 8272 \pm 182$ pc, within 0.14σ and 0.03σ of the GRAVITY interferometric values, at a reduced χ^2 of 0.959. The model reproduces the gravitational-redshift excess at the 2018 pericentre passage ($205\,\mathrm{km\,s^{-1}}$ peak) detected by GRAVITY Collaboration (2018), and the Schwarzschild precession of 11.98 arcmin per orbit confirmed by GRAVITY Collaboration (2020). At the GRAVITY mass and distance the simulated photon ring diameter is $49.68\,\mu$as, inside the EHT measurement of $51.8 \pm 2.3\,\mu$as (Event Horizon Telescope Collaboration, 2022). Freeing the quadrupole moment in a Hartle–Thorne extension yields $\varepsilon_q = 0.00 \pm 0.93 \times 10^5$ (68 per cent), consistent with the no-hair theorem.

This report is the long-form account of that programme. Part I is a plain-language primer for readers without a physics background. Parts II–IV derive, from first principles, every formula the simulator implements; document the numerical methods and their error analysis; present the full results with their statistical treatment; and record the complete development history, test suite, and validation gates through which the instrument was built.

Notation and Conventions

Geometric units $G = c = 1$ are used throughout unless stated; masses, lengths and times are quoted in units of the gravitational radius M (for Sgr A*: $GM/c^2 = 5.1243^{-1}$ scaling on the sky, $GM/c^3 = 21.17$ s). Metric signature $(-,+,+,+)$; Boyer–Lindquist coordinates (t, r, θ, ϕ).

Reproducibility Statement

Every quantitative statement in this report that is not a cited literature value is typeset from a macro in `numbers.tex`, generated by `scripts/make_numbers.py` from machine-readable artefacts produced by the simulation; every figure is generated by `make figures` (matplotlib) from the same artefacts. The single command `make reproduce` re-runs the full chain — unit tests, reference renders, sub-ring metrology, orbit fits, calibration, sweeps, and figure generation — from source. A number that cannot survive this cycle cannot appear in this document.

Part I

Foundations for the Non-Specialist

Chapter 1

Gravity, Mass, and the Idea of a Black Hole

[Chapter under construction — scope: escape velocity, the Schwarzschild radius, horizons, singularities, and a sense of scale. See `paper/THESIS_PLAN.md`.]

Chapter 2

Spacetime and Einstein's Gravity, Gently

[Chapter under construction — scope: curved spacetime, geodesics, light bending, time dilation, and what a metric is. See `paper/THESIS_PLAN.md`.]

Chapter 3

Spinning Black Holes and the Shadow

[Chapter under construction — scope: Kerr vs Schwarzschild, frame dragging, the ergosphere, the photon sphere, and how the shadow forms. See `paper/THESIS_PLAN.md`.]

Chapter 4

Light, Heat, and Colour: Reading the Disk

[Chapter under construction — scope: blackbody colour, the accretion disk, Doppler beaming, and the three redshifts. See `paper/THESIS_PLAN.md`*.]*

Chapter 5

Orbits, S-Stars, and Weighing a Black Hole

[Chapter under construction — scope: Kepler's ellipse, astrometry vs radial velocity, precession, and how mass and distance emerge. See `paper/THESIS_PLAN.md`.]

Chapter 6

The Toolkit in Plain Words

[Chapter under construction — scope: geometric units, tensors gently, chi-squared and confidence intervals, MCMC, and the no-hair theorem in one picture. See `paper/THESIS_PLAN.md`.]

Part II

Theory and Methods

Chapter 7

Introduction and Strong-Field Context

[Chapter under construction — scope: Sgr A as the unique dual laboratory; historical thread; contributions; reader's guide. See* `paper/THESIS_PLAN.md`*.]*

Chapter 8

Kerr Geometry from First Principles

[Chapter under construction — scope: the metric in two forms, Killing structure, characteristic surfaces, photon orbits, the BPT ISCO, and circular-orbit energetics. See `paper/THESIS_PLAN.md`*.]*

Chapter 9

Geodesics and the Carter Constant

[Chapter under construction — scope: Hamilton-Jacobi separation, the potentials R and Theta, Mino time, and the second-order formulation. See `paper/THESIS_PLAN.md`*.]*

Chapter 10

Numerical Methods and Error Analysis

[Chapter under construction — scope: Dormand-Prince 5(4), step control, constraint projection, complex-step differentiation, convergence studies. See `paper/THESIS_PLAN.md`.]

Part III

Results

Chapter 11

The Reference Ray Tracer, Shadow, and Critical Curve

[Chapter under construction — scope: ZAMO camera, backward tracing, the Synge formula, the Bardeen critical curve, spherical photon orbits and Lyapunov exponents, gravitational lensing of the sky. See `paper/THESIS_PLAN.md`*.]*

Chapter 12

The Accretion Disk: Novikov-Thorne and Page-Thorne

[Chapter under construction — scope: thin-disk conservation laws, the full flux integral, the redshift factor, g^4 beaming, blackbody colour, and the spin-morphology of the disk image. See `paper/THESIS_PLAN.md`.]

Chapter 13

Photon-Ring Substructure

[Chapter under construction — scope: sub-ring orders, demagnification, measured flux ratios and widths, the axial image tower, detectability forecasts. See `paper/THESIS_PLAN.md`*.]*

Chapter 14

Timelike Orbits and the S-Star Forward Model

[Chapter under construction — scope: Darwin elements, sky projection, the Romer delay, and the redshift decomposition. See `paper/THESIS_PLAN.md`.]

Chapter 15

Orbit Fitting, Statistics, and the S2 Results

[Chapter under construction — scope: the chi-squared objective, least squares, covariance, MCMC posteriors, and the S2 results against GRAVITY. See `paper/THESIS_PLAN.md`.]

Chapter 16

The Wider S-Star Cluster: A Joint Mass Fit

[Chapter under construction — scope: multi-star data pipeline, the shared-potential model, the joint mass posterior. See `paper/THESIS_PLAN.md`.]

Chapter 17

A No-Hair Test with a Free Quadrupole

[Chapter under construction — scope: multipole moments, the Hartle-Thorne/quasi-Kerr metric, the quadrupole precession, and the ε_q constraint. See `paper/THESIS_PLAN.md`.]

Part IV

Engineering, Validation, and Closing

Chapter 18

GPU Implementation and Validation

[Chapter under construction — scope: the bindless compute pipeline, double precision on GPU, banded dispatch, and pixel-level validation. See `paper/THESIS_PLAN.md`.]

Chapter 19

Sagittarius A* Calibration and Parameter Sweeps

[Chapter under construction — scope: the angular-scale conversion, the ring-diameter measurement, and the spin-inclination sweep. See `paper/THESIS_PLAN.md`.]

Chapter 20

The Interactive Instrument

[Chapter under construction — scope: the engine, presets, the cinematic menu, and the display-only turbulence layer - separated from the physics. See `paper/THESIS_PLAN.md`*.]*

Chapter 21

Development History, Iterations, and the Test Suite

[Chapter under construction — scope: the twelve validation gates, the complete test suite with tolerances, and the bug-hunt narratives. See `paper/THESIS_PLAN.md`*.]*

Chapter 22

Discussion and Limitations

[Chapter under construction — scope: idealisations, degeneracies, and comparison with current and future instruments. See `paper/THESIS_PLAN.md`.]

Chapter 23

Conclusions and Future Work

[Chapter under construction — scope: verified findings and the roadmap. See `paper/THESIS_PLAN.md`.]

Appendix A

Extended Algebra

[Chapter under construction — scope: full derivations too long for the main text: BPT ISCO, Page-Thorne integral, Hartle-Thorne expansions. See `paper/THESIS_PLAN.md`*.]*

Appendix B

Algorithm Pseudocode

[Chapter under construction — scope: integrator, constraint projection, crossing bisection, fitter, sampler. See `paper/THESIS_PLAN.md`*.]*

Appendix C

Number Provenance and Verification Gates

[Chapter under construction — scope: the artefact table for every number and the gate table with measured results. See `paper/THESIS_PLAN.md`.]

Appendix D

The Reproduction Pipeline

[Chapter under construction — scope: what make reproduce runs, step by step. See `paper/THESIS_PLAN.md`.]

Appendix E

Notation Glossary

[Chapter under construction — scope: every symbol used in this report, with its first-use location. See `paper/THESIS_PLAN.md`*.]*

Bibliography

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GRAVITY Collaboration 2018, Astronomy and Astrophysics, 615, L15

GRAVITY Collaboration 2020, Astronomy and Astrophysics, 636, L5

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